

The Engram as Hash, the Readout as Separation Operator: Testing Two Recent Theories of Content-Addressable Memory

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Abstract

We report two direct empirical tests of recent theoretical claims about content-addressable, hippocampal-style memory, using a controlled testbed in which solved Sudoku grids are stored as sparse binary engrams: each grid’s 81×81 same-digit coincidence matrix is projected through a fixed random sparse matrix and sparsified by k -winner-take-all, then retrieved through an attention-style key-value readout. First, we test whether this encoder – structurally identical to the random-projection-plus-winner-take-all circuit identified in the fly olfactory mushroom body – actually behaves as a similarity-preserving hash, rather than assuming the analogy by construction. Across 390 systematically corrupted queries spanning the full corruption range, engram overlap with the true stored code tracks input similarity almost monotonically (Spearman $\rho = 0.974$, $p < 10^{-250}$), and the floor of that relationship lands almost exactly on the chance overlap predicted from code geometry alone (K/H), ruling out a leftover-correlation artifact. Second, we test a very recent proposal that classical correlation-matrix memory, sparse distributed memory, dense associative memory, and Transformer-style attention are the same key-value computation, differing only in a “separation operator” nonlinearity applied to key-query similarity, by swapping our readout’s operator across all four families under a fully paired noise sweep on the identical stored engrams and identical corrupted queries. The results both confirm and refine the unification: some separating nonlinearity is a strict requirement for retrieval to function at all among many competing memories (linear reweighting fails even with a noiseless query), but robustness does not track how aggressively an operator discriminates – it tracks whether the operator sharpens by an absolute similarity *gap* or a similarity *ratio*. Ratio-based sharpening (a rectified-power nonlinearity, the dense-associative-memory operator) collapses at roughly half the noise level that gap-based sharpening (softmax) tolerates, despite matched zero-noise selectivity, an effect we trace to an extreme-value calculation over the $M - 1$ competing stored memories: a power of a similarity ratio loses discriminative weight far faster than an exponential of the corresponding absolute gap does, once a non-negligible number of competitors guarantees an interference floor. Both results are obtained on the same fixed encoder instantiation and the same $M = 200$ stored engrams, so together they characterize, on one implemented system, both stages – encoding and readout – of a hippocampal-style key-value memory. All code, data, and reproduction instructions accompany this paper.

1 Introduction

Biological memory systems, and the hippocampus in particular, are widely understood to store information as sparse, combinatorial codes and to retrieve it through content-addressable, key-value-like association rather than by address [1, 2, 3]. Two specific, independently motivated theoretical claims about this kind of architecture have particular purchase right now, and both are directly testable in an implemented system rather than only arguable by analogy.

The first claim concerns the *encoder*. Projecting a high-dimensional input through a fixed random matrix and sparsifying with winner-take-all is a generic combinatorial coding scheme, in the spirit of Willshaw associative memory [4]. The same computation – random projection followed by winner-take-all – has been identified as the circuit motif of the fly olfactory mushroom body, where it is understood not merely as sparsification but as a *locality-sensitive hash*: a transformation under which nearby inputs remain nearby and distant inputs remain distant, in the specific technical sense used for approximate nearest-neighbor search [12]. Architectures that reuse this same random-projection-plus-winner-take-all pattern elsewhere are routinely described as hash-like by analogy to that result. We treat this as a claim to check rather than assume: an encoder can sparsify a representation into a fixed-size combinatorial code without preserving similarity structure at all – a pure randomizer would still produce a legitimate-looking sparse code – so the analogy needs verification for each specific application, not inheritance from the motif alone.

The second claim concerns the *readout*. Gershman, Fiete, and Irie recently proposed that four architectures usually treated as distinct in the memory literature – classical correlation-matrix memory [8], sparse distributed memory [9], (modern) dense associative memory [5, 6], and Transformer-style self-attention [10, 11] – are the same key-value computation, differing only in the nonlinearity (“separation operator”) applied to key-query similarity before it weights the retrieved values [7]. This is a strong, checkable claim: if it is right, one implemented key-value memory should be able to instantiate all four architectures simply by swapping this one nonlinearity, and the theory further predicts how they should differ in behavior – sharper separation should trade robustness for precision.

We test both claims directly in a single implemented system: a hippocampal-inspired key-value memory built to store and retrieve solved Sudoku grids. Sudoku gives a convenient testbed because a solved grid’s relational structure (which cells share a digit) is a well-defined combinatorial object with an unambiguous ground truth, and because query corruption – flipping engram bits, or corrupting the input relation itself – is easy to control and sweep exhaustively. Neither test below depends on Sudoku specifically: both probe general properties of the encoder and the readout that would transfer to any content stored the same way. What Sudoku buys us is a controlled, fully paired, exactly reproducible sweep for each, run on the *same* fixed encoder and the *same* $M = 200$ stored engrams, so that the two results characterize one coherent system rather than two unrelated experiments.

2 Model

A solved Sudoku grid g is an assignment of digits $\{1, \dots, 9\}$ to $N = 81$ cells. Its *coincidence matrix* $C \in \{0, 1\}^{N \times N}$ records which cell pairs share a digit: $C_{ij} = \mathbb{I}[g_i = g_j]$ for $i \neq j$. This matrix, flattened, is the input to the encoder.

A fixed random sparse matrix $W_{SH} \in \mathbb{R}^{H \times N^2}$ (density ρ , nonzero entries drawn i.i.d. from $\mathcal{N}(0, 1)$) projects the flattened coincidence matrix into an H -dimensional layer. The *engram* is the top- K k -winner-take-all binarization of this projection:

$$h = \text{topK}(W_{SH} \text{vec}(C)) \in \{0, 1\}^H. \quad (1)$$

Every stored grid also carries a *value* vector $b \in \mathbb{R}^N$, a per-cell digit-to-drive-level mapping.

Given M stored (key, value) pairs $\{(h_m, b_m)\}_{m=1}^M$ and a query engram h_q , retrieval first computes a similarity score against every stored key, $s_m = h_m \cdot h_q$ (the shared-active-unit overlap between

query and stored engram m), then returns a weighted combination of stored values,

$$\hat{b} = \sum_{m=1}^M p_m b_m, \quad p = \sigma(s), \quad (2)$$

where σ , the *separation operator*, maps the vector of raw similarities s to a set of non-negative retrieval weights. The reference configuration used throughout Section 3.1 and as the baseline in Section 3.2 takes σ to be softmax with temperature τ , $p_m \propto \exp(s_m/\tau)$: a pure content-addressable lookup with no learned parameters beyond the fixed encoder. Section 3.2 swaps σ for five further operators, described there.

Standard parameters throughout this paper: $H = 1000$ hippocampal units, $K = 150$ active units per engram (density $K/H = 0.15$), and projection density $\rho = 0.02$. Both experiments below reuse the identical fixed W_{SH} and the identical $M = 200$ stored engrams (same random seed), so they characterize one specific encoder-plus-readout instantiation rather than two separately tuned systems.

3 Results

3.1 The engram code as a similarity-preserving hash

Neither the encoder’s design (Section 2) nor any downstream decode-accuracy curve directly tests whether $\text{topK}(W_{SH} \text{vec}(C))$ preserves relative similarity between inputs – the specific property that random projection followed by winner-take-all is known to provide in the fly olfactory mushroom body [12]. Decode accuracy measures a downstream consequence of that property, if it holds, mixed together with whatever the readout separately contributes; it cannot distinguish “the encoder preserves similarity” from “the readout is robust enough to tolerate an encoder that does not.” We test the encoder’s own geometry directly.

For 30 stored grids sampled from the $M = 200$ pool, we corrupted each grid’s true coincidence matrix C_{true} by flipping a controlled fraction of its (symmetric) same-digit entries, swept from 0% to 100% across 13 levels, and measured two quantities on each corrupted instance: the *input similarity* to the true grid (Jaccard overlap of the same-digit pair sets between C_{noisy} and C_{true}) and the *engram similarity* of the resulting codes ($|h_{\text{noisy}} \cap h_{\text{true}}|/K$). Across the resulting 390 corrupted instances, the two are strongly and monotonically related (Spearman $\rho = 0.974$, $p < 10^{-250}$; Figure 1, left) – direct confirmation that the encoder behaves as a similarity-preserving hash rather than merely as a randomizing sparsifier. The floor this relationship approaches at full corruption (0.149 mean engram overlap at 0% input similarity; Figure 1, right) matches the chance overlap expected between two independent K -sparse codes in an H -dimensional space, $K/H = 150/1000 = 0.15$, almost exactly – confirming that residual overlap at maximal corruption reflects code geometry, not a leftover correlation the corruption procedure failed to remove.

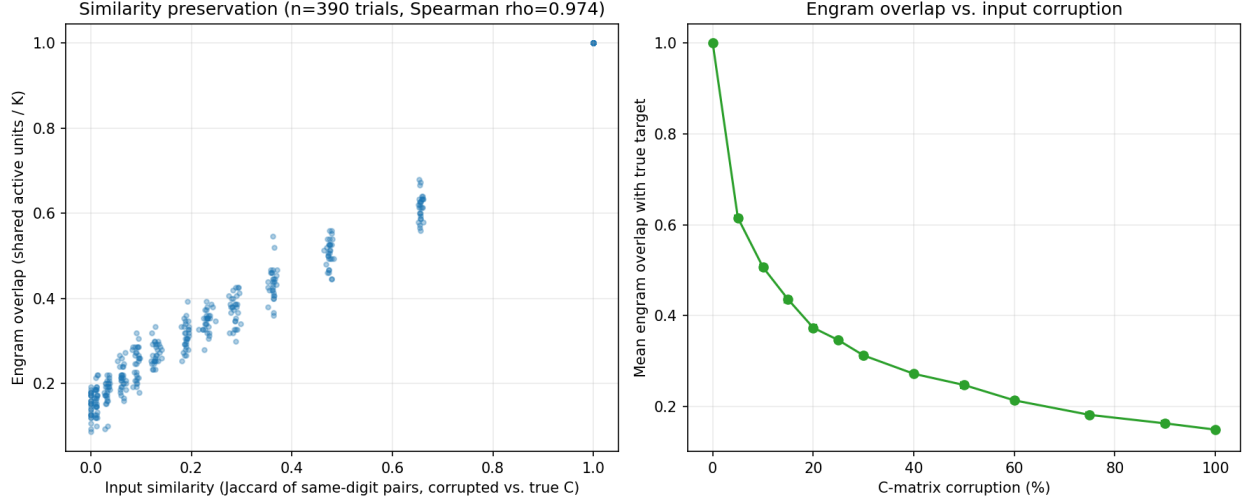


Figure 1. Left: engram overlap vs. input (coincidence-matrix) similarity, pooled across all 13 corruption levels and all 30 target grids ($n = 390$). Right: mean engram overlap with the true target vs. percent corruption of the input coincidence matrix (error bars: standard error across the 30 targets at each level).

This result matters beyond confirming an analogy. A similarity-preserving encoder is exactly what a downstream content-addressable readout needs in order to degrade gracefully rather than catastrophically under query noise: if two inputs that are 90% similar could map to completely uncorrelated codes, no readout built on top of the code could recover a graceful noise-tolerance curve, however good the readout’s own separation mechanism is. The next section holds this encoder fixed and asks how much of the resulting system’s robustness comes from the readout side.

3.2 Separation operators: gap-based vs. ratio-based sharpening

Gershman, Fiete & Irie [7] note that softmax, a hard top- j threshold, and a rectified-power nonlinearity are, respectively, the separation operators of Transformer-style attention, sparse distributed memory [9], and dense associative memory [5, 6] – three architectures differing only in this one nonlinearity, alongside plain (identity) correlation-matrix memory [8], which applies no separation at all. We test this unification directly on the encoder characterized above: using the same fixed W_{SH} and the same $M = 200$ stored engrams as Section 3.1, we corrupt each query by flipping a swept fraction of its engram bits, decode digit-cell values from $\hat{b}$ for each of six separation operators evaluated on the identical corrupted queries at every noise level (a fully paired design, 100 trials per noise level, no error-correction cleanup pass, so as to isolate each operator’s own behavior rather than a downstream correction mechanism’s).

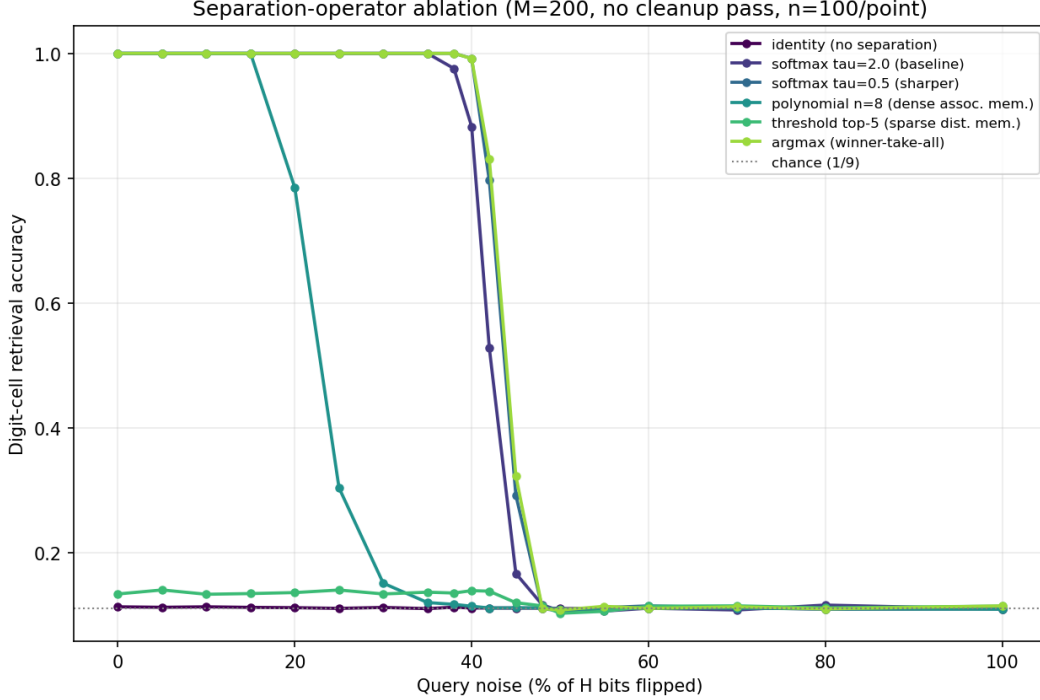


Figure 2. Digit-cell retrieval accuracy vs. query noise for six separation operators, evaluated on identical corrupted queries at every noise level ($M=200$, $n=100/\text{point}$, no cleanup pass).

Two findings depart from the intuition that a separation operator’s robustness is set by how aggressively it discriminates (“sharp separation is powerful but fragile” [7]). First, identity separation – no nonlinearity at all – performs at chance *even with a perfectly clean query* (11.4% vs. 100% for softmax at 0% noise): with $M=200$ competing memories, a linear reweighting cannot overcome the combined pull of the other 199 stored values on the weighted average, so some separating nonlinearity is not merely a robustness refinement here but a strict requirement for retrieval to function at all. Second, and more surprisingly, sharper is not uniformly more fragile: a hard-threshold top-5 average (naively implementing sparse distributed memory’s equal-weight readout) is *also* near chance at every noise level, including zero, because averaging five equally-weighted stored values – one correct, four spurious – dilutes the correct digit rather than recovering it, unlike real sparse distributed memory’s per-bit majority vote over redundant patterns. Meanwhile argmax (winner-take-all, the “ideal noiseless” operator [7]) and a sharpened softmax ($\tau = 0.5$) hold up *at least as well as* the baseline softmax ($\tau = 2.0$) right up to a shared collapse near 45–48% noise – sharpness itself is not the fault line.

The real fault line is gap-based vs. ratio-based sharpening. The rectified-polynomial operator (dense-associative-memory-style, exponent $n = 8$) collapses far earlier, around 20–25% noise, despite normalized zero-noise selectivity comparable to the other sharp operators. We traced this mechanistically, with a dedicated exhaustive calculation over all $M = 200$ stored items (Table 1): even at zero noise, the best-competing wrong memory already reaches, on average, similarity within a factor of 3.39 of the true match’s – an extreme-value effect of having 199 competitors, well below the naive factor of 6.67 predicted from the mean chance-overlap floor alone (H/K) – and this ratio shrinks toward 1 as query noise grows.

Table 1. Mean competitor ratio (true-match similarity / best-wrong-match similarity), exhaustive over all $M=200$ stored items, vs. query noise.

Query noise	0%	20%	30%
Mean competitor ratio	3.39	1.92	1.45

(Naive factor from the mean chance-overlap floor alone, H/K : 6.67.)

A power of a ratio shrinking toward 1 loses discriminative weight far faster than an exponential of the corresponding absolute gap does, for the same starting margin – softmax’s exponential-of-a-gap form is intrinsically more robust than a power-of-a-ratio form once a sizable number of competing memories guarantees a non-negligible interference floor. This is a refinement, not a contradiction, of the unification in [7]: the four architectures are indeed the same computation with a different nonlinearity plugged in, but which nonlinearity is the right engineering choice depends on the similarity statistics of the memory system it operates over – specifically, how many competing memories are stored and how much residual similarity they carry – not on sharpness alone.

4 Discussion

Taken together, these two results characterize both halves of a hippocampal-style key-value memory – the encoder and the readout – on the same implemented system, rather than testing either theoretical claim in isolation. Section 3.1 shows that the encoder half genuinely earns its analogy to the fly olfactory hashing circuit: it is not merely sparse and combinatorial, but approximately similarity-preserving, the specific geometric property a content-addressable readout depends on. Section 3.2 shows that the readout half genuinely instantiates the unification proposed by [7] – the same stored engrams and the same encoder support correlation-matrix-memory-style, sparse-distributed-memory-style, dense-associative-memory-style, and Transformer-attention-style retrieval simply by swapping one nonlinearity – while also sharpening that theory’s own stated intuition. The theory’s framing (sharper separation trades robustness for precision) predicts a single ordered axis; what we find instead is two families that do not order along that axis at all: gap-based operators (softmax at either temperature, argmax) remain close to tied with each other regardless of sharpness, while the one ratio-based operator tested (rectified power) collapses at roughly half their noise tolerance despite comparable zero-noise selectivity. The mechanism we identify – an extreme-value competitor-ratio calculation over $M - 1$ stored memories – is a property of *any* memory system with enough stored items to guarantee a non-negligible interference floor, not an artifact of the Sudoku testbed, so we expect the same gap-vs-ratio distinction to reproduce in other implementations of this same operator family.

A natural question is whether the two results are related beyond sharing an experimental platform: does a similarity-preserving encoder make ratio-based sharpening fail sooner, or is this purely a readout-side effect? Our design cannot separate these because both experiments share the identical encoder; testing whether the gap-vs-ratio distinction depends on the encoder’s similarity-preservation would require deliberately degrading that property (e.g. by adding an explicit decorrelating nonlinearity to W_{SH} ’s output before k -WTA) and re-running the separation-operator sweep on the degraded code – a direct extension of this work that we have not pursued here.

5 Methods

Encoder. $H = 1000$, $K = 150$ (density $\rho_K = K/H = 0.15$), fixed random projection density $\rho = 0.02$, nonzero entries i.i.d. $\mathcal{N}(0, 1)$. Both experiments use the identical fixed W_{SH} and the

identical pool of $M = 200$ stored Sudoku grids (same random seed for grid generation, projection, and engram construction).

Section 3.1 (hash test). 30 grids sampled without replacement from the $M = 200$ pool. Corruption: symmetric bit flips on the coincidence matrix’s off-diagonal same-digit entries, at fractions $\{0, 5, 10, 15, 20, 25, 30, 40, 50, 60, 75, 90, 100\}\%$ of all $\binom{81}{2}$ cell pairs, independently sampled per trial. Input similarity: Jaccard overlap of the same-digit pair sets between corrupted and true coincidence matrices. Engram similarity: shared active units between corrupted and true engrams, divided by K . Statistic: Spearman rank correlation, pooled across all $30 \times 13 = 390$ (target, corruption-level) pairs.

Section 3.2 (separation-operator ablation). Query corruption: a swept fraction of the $H = 1000$ engram bits flipped uniformly at random (independent draw per trial). Six separation operators evaluated on identical corrupted queries at every noise level: identity (no separation), softmax at $\tau = 2.0$ (reference) and $\tau = 0.5$ (sharpened), argmax (winner-take-all), hard top-5 uniform average, and rectified power (exponent $n = 8$, normalized by K). No conflict-mask cleanup pass is applied, isolating each operator’s raw behavior. 100 trials per noise level per operator, identical corrupted queries reused across operators within a trial (paired design). The competitor-ratio calculation (Table 1) is exhaustive rather than sampled: for each of the $M = 200$ stored items and each of three noise levels, the true-match similarity and the maximum similarity among the other 199 stored items are computed directly and averaged.

Code and Data Availability

All simulation code, raw data (CSV), and figure-generation scripts are available at <https://github.com/ghub-res1578/sudoku-hippocampal-key-value-memory>. This paper draws specifically on `hash_similarity_preservation.py` (Section 3.1), `separation_operator_ablation.py` (Section 3.2), `separation_operator_competitor_ratio.py` (Table 1), and the shared core module `hippocampus_drive_readout.py` that both scripts import the encoder and stored engram pool from. The repository’s README gives exact commands to regenerate every number, table, and figure above.

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